

Aims 1 and 2 update:

measuring the language component in multilingual LLMs

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17 models · 122–128 languages · 31 pre-registered predictions · 4/4 blind prospective test

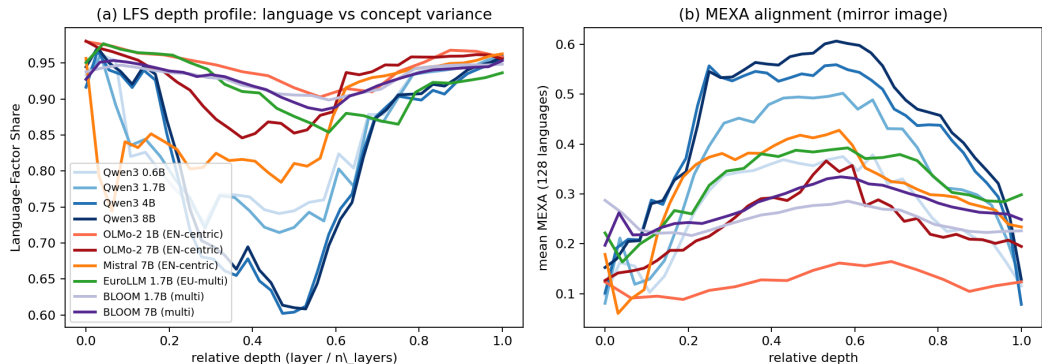
The question, in the proposal's own words

*"It is **not clear how strong the language component** in the representation is, or should be..."* (§3.1.1)

*"At this point, **we have not singled out a specific metric.**"* (§3.1.4)

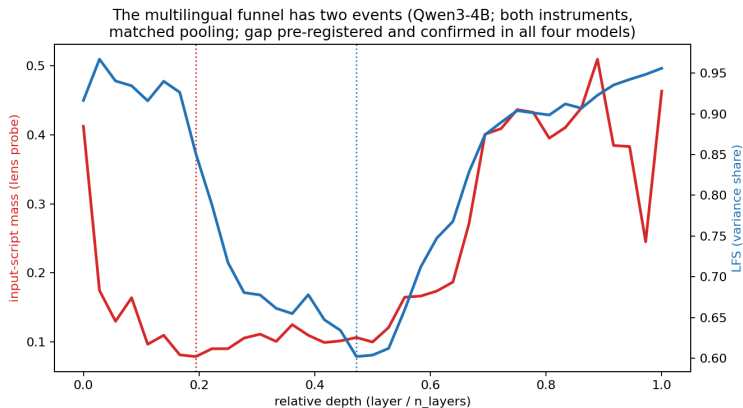
This talk: that metric, built, validated, stress-tested.

The answer: the Language-Factor Share (LFS)



Per-layer variance share attributable to *language identity vs meaning*. Every model funnels; the depth varies 0.04–0.34: **a measurable, model-dependent quantity**. Family moves it more than scale (this grid).

Your three-stage sketch has finer structure: two events



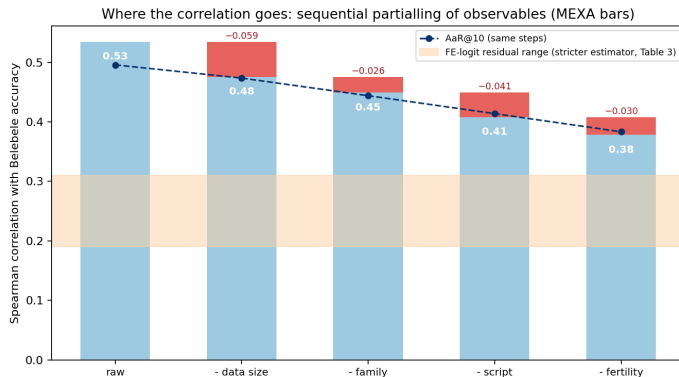
Script is abandoned *early*; concepts converge *mid*. Pre-registered, confirmed twice, under matched pooling.

Before the findings: how we checked our own results

- ▶ My first hub estimate: latent-over-English $+1.04$. **I found the confound myself**; the corrected number is $+0.08$; still 113 to 124 of 127 languages by model.
- ▶ **31 pre-registered predictions, 21 passed; every failure documented** next to every success (several failures became new findings).
- ▶ **7 self-caught errors** with mechanisms in the report's ledger: a collapsed estimator withdrawn, a silently-unfair eval protocol caught, precision corruption, a gamed loss detected by the suite itself.

The claims that follow survived all of this.

On our grid, the standard validation number is roughly half confound



Data-rich languages score high on *both* alignment and tasks. **Confounder-adjusted validity:** partial out tokens, family, script, tokenizer; what survives is the metric's real signal.

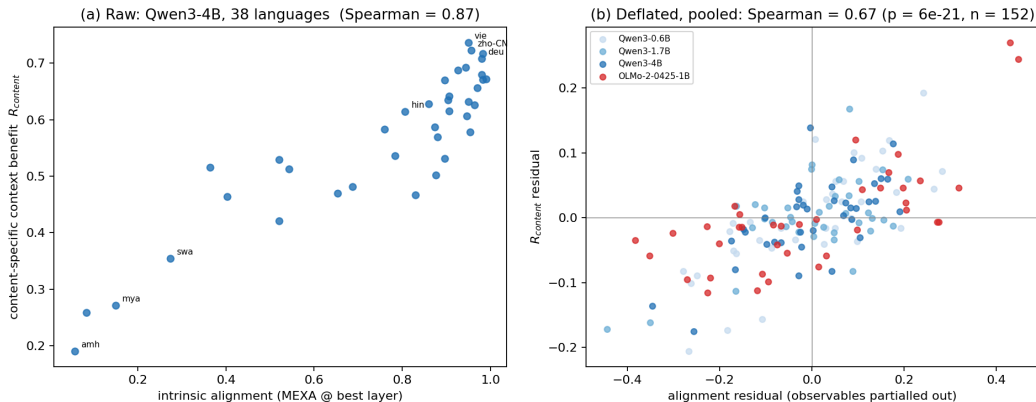
Confounder-adjusted results

	vs. raw accuracy	vs. transfer residual
AaR@10 (tail measure)	0.50	0.23–0.31
MEXA (reference)	0.53	0.19–0.27

Estimator range $\rho \approx 0.19\text{--}0.31$; the July figure of about 0.59 on a benchmark-capable subset is withdrawn (not reproducible from any stored artifact). **The measures are not ranked against each other; AaR's job is the failing tails that means hide.** MEXA is shown for reference.

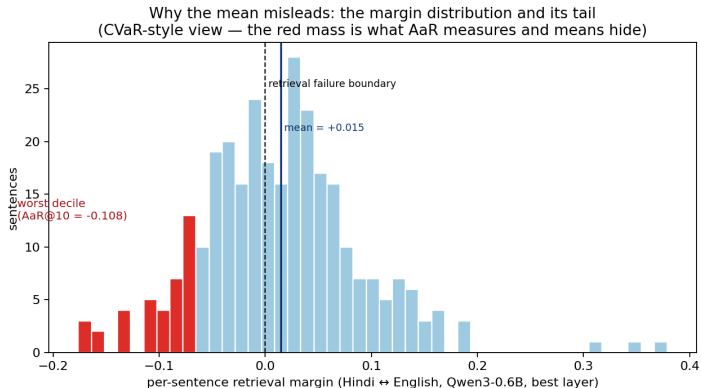
Preregistered prediction P4 passed on one held-out model and is not used as a ranking.

Shared-space alignment predicts behavior, beyond observables



Alignment at the LFS-identified layer predicts how much the model *uses* a foreign-language context in generation: confounder-adjusted $\rho = 0.67$ (clustered CI [0.55, 0.77]; wrong-document control rules out entity leakage). **Observational.** Tested under intervention (Aim 2), the association does not transfer.

Why tails: the mean misleads



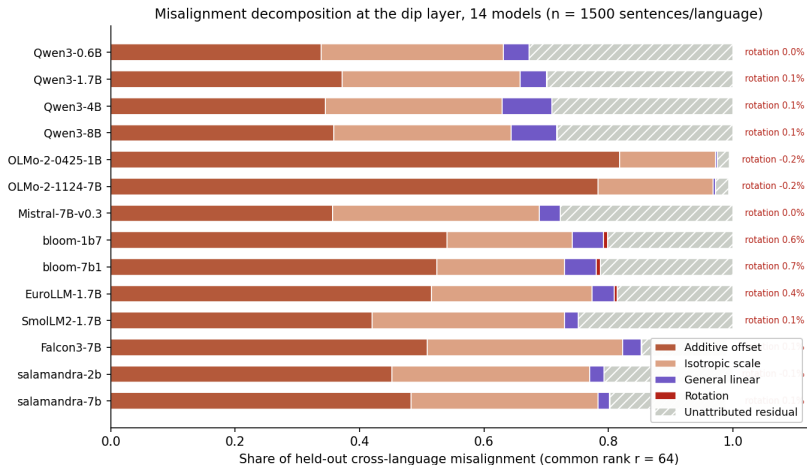
Hindi: mean margin $+0.015$ (looks fine); worst decile -0.108 (failing). At $n=1500$: **40–50% of languages in the best models are mean-positive, tail-negative** (Qwen3-8B: Turkish $+0.13$ / -0.26). The tail is *semantic* (anaphora and culture-bound vocabulary), not entity noise. We read it.

Generalization: pre-registered prospective test, 4/4 blind

Prediction (frozen before running)	Result
Unseen families match their recipe class (SmolLM2 shallow, Falcon3 intermediate)	PASS
Attribution criterion holds at 12 models	PASS
No validity collapse on unseen families (held-out <i>exceeded</i> dev CI)	PASS
Benchmark-capable Δ reproduces (retraction pre-committed)	PASS

The design-circularity objection is substantially narrowed: the suite classified two unseen families **under blind scoring** (one benchmark-capable).

We attacked the metric with calibrated faults, then measured its blind spots



Real cross-language misalignment is offset + scale; **rotation is at most 0.7%** (14 models, $n=1500$; injected rotations recovered exactly, so this is measured absence). The one attack that is missed by every alignment metric tested (uniform collapse) is caught only by our

Aim 2: 28 conditions, and a defect in our own summary

- ▶ **The objective §3.2.2 specifies degenerates by 89%.** A squared difference between matched word representations is minimized at $h = 0$. Pre-registered, and it happened at all three seeds.
- ▶ **Neither LFS nor MEXA detects it.** Both are scale-invariant. The scale-collapsed condition posts the *best* retrieval and *best* tail alignment in the study; the sound condition gains nothing.
- ▶ **The components separate them by 20×.** The ratio $V_{\text{lang}}/(V_{\text{lang}}+V_{\text{concept}})$ divides out exactly the quantity that failed.
- ▶ **A collapse guard is necessary and not sufficient:** bounded hinge (0.81) against an unbounded objective (600). A 1000× weight sweep saturates.

Report the profile (V_{lang} , V_{concept} , **resid**, $\|h\|$), **not the share.** It catches every failure in this study; the share catches none.

Three things I'd value your input on:

1. Where to go after the Aim-2 study: five objectives failed, and the collapse guard was necessary but not sufficient. Design the next one together?
2. Venue, and ordering of pre-submission items (native-source replication, exact-count subset, FLORES).
3. Coordination with adjacent work in the group.

Boundaries, named: sentence-level, translated-news domain, one full scaling family, proxy token counts, additive-share construct, and observational validity that does not transfer under intervention. The last is now *measured*: rotation is 0.06–0.2% of real misalignment, and the collapse attack that is missed by every alignment metric is caught by our variance-preservation check. The point of this work is not to hide uncertainty; it is to measure inside it.

Full technical report (methods, statistics, and error ledger) and release-ready code available.